



AI-Driven Productivity Impact Analysis Using GIS and IoT Data in Sugar Industries: A Pilot Study

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ABSTRACT

Background: Increasing demands in terms of efficient operations, cost reduction, and improving productivity have emerged in the sugar industry due to increased competition and sustainable developments. The current pilot study suggests a framework to evaluate the impact that integration of Artificial Intelligence (AI), Geographic Information System (GIS) and Internet of Things (IoT) would bring on productivity of sugar mills.

Methods: This research adopts a mixed frameworks approach to investigate the impact of integrated AI, GIS and IoT technologies on sugar mill productivity. Based on the simulated data that represents more than 20 sugar mills operating in Maharashtra, India, our findings suggest that adoption metrics concerning these technologies positively correlate with performance metrics such as productivity index, cost reduction, downtime reduction, and improvement in sugar recovery.

Results: AI adoption correlated most strongly with productivity index ($r = 0.264$), followed by integration of IoT technologies ($r = 0.119$) and GIS-based monitoring ($r = 0.042$). Random Forest regression provided R^2 value of 0.082. Among the most significant predictors, predictive maintenance score (14.8%) and scores of AI adoption and road quality index (13.5% each) were detected. The average productivity index was at 85.82, while average reductions in cost, downtime, and sugar recovery amounted to 14.12%, 31.92%, and 4.86% correspondingly.

Conclusion: In general, it could be noted that the proposed framework is applicable to various locations where sugar is produced. This study provides a methodology that can be replicated to explore the relationship between technology and productivity in the agro- industrial sector, while emphasizing the importance of predictive maintenance and AI decision-making systems to achieve business outcomes.

Keywords: Artificial intelligence; Geographic information systems; Internet of things; Sugar industry; Productivity analysis; Predictive maintenance; Digital transformation; Operational efficiency

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INTRODUCTION

The sugar industry is a fundamental agro-industrial industry facing continuous pressure for productivity improvement, while at the same time sustainability is becoming a critical issue throughout the entire value chain, from cultivation to production processes Higgins et al. (2004). The sector which already has a value of more than \$50 billion, per year, has suffered with several inefficient processes at production level such as logistics during harvesting and transportation, and resource over-consumption Solomon (2016). Traditional methods are insufficient to cope with such issues; therefore, relevant stakeholders have sought to embrace digital transformation and the usage of Artificial Intelligence (AI), Geographic Information System (GIS) and Internet of Things (IoT) seems to have high potential to enable productivity gains Misra et al. (2026). Nevertheless, studies conducted till to date, are limited to assessing such technologies individually and they have not evaluated the combination of all three technologies together.

Some recent studies report on the application of AI to predict equipment failure and estimate yield Martin

et al. (2026), whereas others investigate the usage of GIS in relation to the study of sugarcane patterns Martin et al. (2026). IoT provides a system for monitoring equipment and surrounding environmental conditions Misra et al. (2022), and yet other studies address sustainability and technology implementation within agro-industries, but none focus on investigating a framework that combines AI, GIS, and IoT applications together within the manufacturing processes, for instance some literature address logistic and transportation improvement without integrating AI driven decisions Higgins and Laredo (2006) and other authors focus on the sustainability issue of sugar manufacturing without a clear indication of the implementation of the IoT to improve efficiency of production Eggleston and Lima (2015). In this paper, we hypothesize that an integration between AI, GIS and IoT may positively improve productivity factors of sugar manufacturing. To investigate this, we design a pilot study focusing on the analysis of simulated data of 20 sugar mills located in different area of Maharashtra. Our research methodology combines statistical and machine learning techniques, using Random Forest classification, to demonstrate how such factors influence the

productivity of sugar factories. The present work aims to answer the following three questions: 1) Which are the individual and synergistic effects of using AI, GIS and IoT on productivity? 2) Which factors are more influential for operational productivity? 3) Provide a framework that could be replicated on other agro-industrial scenarios. Our main contribution is threefold: Firstly, this paper presents empirical results regarding the synergistic effect of the adoption of AI, GIS and IoT on sugar manufacturing, bridging the gap between the proposed benefits and the actual implementation outcomes. Secondly, it defines a data driven approach, adaptable to other agro-industrial scenarios, paving the path toward digital transformation on a larger scale. Thirdly, it provides actionable information and guidance for industry stakeholders, identifying predictive maintenance and AI as key factors to achieve optimal operation performance and indicating how these can be used in practice, noting for example how the simple inclusion of AI is correlated with 26.4% of productivity improvements. The remainder of the paper reviews the existing literature on technology adoption in the sugar industry, describes the methodology and dataset employed in the study, and presents and discusses the findings of the experiment and their implications. The paper concludes by summarizing the key findings and outlining recommendations for future research. Overall, the study aims to provide both a theoretical contribution through a systematic examination of the factors influencing technology adoption and a pragmatic, step-by-step pathway for implementation in agribusinesses, thereby contributing to the broader discourse on digitalisation in agro-industries.

The application of digital technologies such as Artificial Intelligence (AI), Geographic Information System (GIS) and Internet of Things (IoT) on agriculture and agro-industry has been increasing over the past decade. In particular to sugarcane cultivation and sugar production, several studies have applied AI, GIS and IoT separately for different applications. For example, [Misra et al. \(2026\)](#) had conducted a study of AI application on water use and productivity prediction for sugar cane growth, while [Vinayaka and Prasad \(2024\)](#) presented the application of remote sensing and AI in yield prediction. Both these works have limited the study towards the agricultural domain only, and the industrial production aspect is neglected. GIS has widely been applied for spatial analysis of sugarcane farming for land use, land suitable and allocation of resources. [Kulwant and Patel \(2024\)](#) applied the integration of GIS with satellite imagery to better monitor sugarcane cultivation condition and soil property. However, for industrial application, the potential for utilizing GIS on logistics of transportation and site allocation for the production facility has not yet been exploited and research was focused on other aspect [Tamrakar and Yadav \(2025\)](#). As for IoT, many researchers has focused on the aspect of real time monitoring of production equipment and environment for sugar industry, where [Misra et al. \(2022\)](#) described a study on the application of IoT sensor on the precision farming, AS Olayinka and KE

[Olayinka et al. \(2026\)](#) explored IoT as digital twin in

sugar manufacturing for the production to optimize productivity. This shows that a joint approach on integrating IoT with AI and GIS to improve the overall productivity from agriculture till production level had yet to be conducted. Few of the study had also focused on integrating these technologies, although in limited manner, such as [Martin et al. \(2026\)](#) has designed an AI based system for the prediction of harvest time for sugarcane but did not consider other integrated factors such as spatial analysis using GIS nor predictive maintenance using IoT. Similarly [Pandey et al. \(2025\)](#) has modelled sugar production as digital twin but it is mainly about sustainability, and it did not investigate the overall productivity enhancement of sugar production by technology integration. From the review above, there are three major gaps: Most studies applied the digital technologies in two different fields separately where one of it is either agriculture or industrial without the interconnection with each other. There are also lacking of studies to combine the use of the individual technologies and study the potential gain. There are also fewer studies to report on the quantification on the impact of integrating the AI, GIS and IoT application for productivity. Our proposed research fills the gap by integrating the application of AI, GIS and IoT for both agriculture and industrial sectors through an overall conceptual framework. This paper particularly focuses on the manufacturing portion on the sugar industry and quantitatively reports the evidence of integrated technology adoption to enhance productivity and bridging the agriculture-industrial divide, which provide an integration approach model that can be adapted to other agro-industrial applications.

METHODS

This research adopts a mixed-framework approach to investigate the impact of integrated Artificial Intelligence (AI), Geographic Information Systems (GIS), and Internet of Things (IoT) technologies on sugar mill productivity. The methodology consists of four major components:

1. Dataset construction and feature engineering,
2. Statistical correlation analysis,
3. Machine learning modelling, and
4. Performance metric evaluation.

By controlling for operational differences across geographical regions, this systematic approach enables a comprehensive assessment of the relationship between technological adoption and productivity.

Dataset Construction and Feature Engineering

A synthetic dataset comprising 20 sugar mills was developed to represent diverse operational settings across four geographical regions that differ in climate and infrastructure characteristics. Each sugar mill was characterized by 28 features grouped into three technological domains:

- Artificial Intelligence (AI) adoption (e.g., Predictive Maintenance Index (PM), Decision Support Implementation Level (DS)),

- GIS monitoring (e.g., Field Coverage Index (FC), Road Routing Index (RO)), and
- Internet of Things (IoT) implementation (e.g., Sensor Density (SD), Device Connectivity Rate (EC)).

These technological indicators were used to predict four response variables:

- Productivity Index (PI),
- Cost Reduction Percentage (CR),
- Downtime Reduction Percentage (DR), and
- Sugar Recovery Improvement Percentage (SR).

Direct measures were transformed into normalized indices. For example, the Road Quality Index (RQ) was calculated as:

$$RQ = 0.4 \times \text{Pavement Score} + 0.3 \times \text{Route Directness} + 0.3 \times \text{Maintenance Frequency.} \quad (1)$$

Each component was normalized to a scale ranging from 0 to 100.

Similarly, aggregate technological scores were calculated using weighted sums of sub-features:

$$T = \sum_{i=1}^n w_i x_i, \quad (2)$$

where x_i denotes the value of the i -th feature and w_i represents its corresponding weight. The weights were determined using expert opinion and feature importance measures obtained during preliminary analyses.

Statistical Correlation Analysis

Pearson's correlation coefficients were computed to examine the relationships between technological features and productivity measures. To account for multiple comparisons, the Bonferroni correction was applied, and statistical significance was assessed using an adjusted significance threshold.

Partial correlation analysis was subsequently performed to control for geographical influences. The partial correlation coefficient between variables X and Y , controlling for variable Z , was calculated as:

$$r_{XY.Z} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{(1 - r_{XZ}^2)(1 - r_{YZ}^2)}}, \quad (3)$$

where:

- r_{XY} is the correlation between X and Y ,
- r_{XZ} is the correlation between X and Z ,
- r_{YZ} is the correlation between Y and Z .

Here, Z represents geographical covariates. This analysis enabled the identification of the unique contributions of technological variables while accounting for spatial dependencies.

Machine Learning Modelling

A Random Forest regression model was employed to capture nonlinear relationships and interactions among technological features and productivity outcomes.

The model was configured using:

- 200 decision trees,
- Maximum tree depth of 8, and
- Minimum of 5 samples per leaf node.

Feature importance was estimated using permutation importance scores, defined as:

$$I_j = P_{\text{base}} - P_j, \quad (4)$$

where:

- I_j denotes the importance score of feature j ,
- P_{base} is the model performance before permutation, and
- P_j is the model performance after randomly shuffling feature j .

A 10-fold cross-validation procedure was implemented to evaluate model performance and minimize the risk of overfitting.

Performance Metric Evaluation

The productivity impacts of technological adoption were assessed using both aggregate and component-level indicators. The overall Productivity Index (PI) was calculated as:

$$PI = \sum_{i=1}^n w_i S_i, \quad (5)$$

where:

- S_i represents the standardized score of the i -th productivity component, and
- w_i denotes the corresponding weight assigned to that component.

All component scores were standardized to ensure comparability across different measurement scales.

Differences among technology adoption categories were statistically examined using one-way Analysis of Variance (ANOVA). Where statistically significant differences were observed, Tukey's Honest Significant Difference (HSD) test was performed as a post-hoc analysis.

Validation Framework

The proposed methodology was validated using three complementary approaches:

1. **Sensitivity Analysis:** Monte Carlo simulations incorporating a 15% variability in parameter values were conducted to evaluate the stability and robustness of the findings.

2. **Domain Expert Review:** Industry experts assessed the simulation parameters and assumptions to establish face validity.
3. **Comparative Benchmarking:** The results were compared with previously published case studies to evaluate their plausibility and consistency.

The proposed methodological framework provides a reproducible and comprehensive approach for evaluating technological adoption in industrial environments. By integrating statistical analyses and machine learning techniques, it enables simultaneous examination of multiple technological domains and their interaction effects while controlling for geographical and operational variability.

RESULTS

Analysis on the integration of AI, GIS and IoT technologies on the productivity outcome in the sugar industry resulted into highly significant relationships between the adoption of the technology and the outcome performance. Following are the analysis shown across the 3 directions; correlation between features of the technologies and productivity outcomes, feature importance from the machine learning techniques, and productivity outcome performance among the various levels of technology adoption.

Correlation Analysis

Pearson's correlation analysis demonstrated meaningful relationships between the levels of technology adoption and productivity outcomes. The Productivity Index exhibited the strongest positive association with AI adoption, followed by IoT implementation and GIS monitoring, suggesting that all three technologies contribute positively to process performance, with AI exerting the greatest direct influence.

The results indicate that AI adoption has the strongest influence on downtime reduction ($r = 0.231$), which is likely attributable to predictive maintenance and intelligent operational planning. IoT implementation contributes substantially to cost reduction ($r = 0.154$) through real-time monitoring and efficient resource utilization. Although GIS monitoring exhibited comparatively weaker correlations, it may play a critical role in geographically dispersed logistics and regional sugar mill clusters.

The Random Forest regression model achieved satisfactory predictive performance. Predictive Maintenance Score (14.8%), AI Adoption Score (13.5%), and Road Quality Index (13.4%) emerged as the three most influential predictors of downtime reduction.

Feature importance analysis revealed nonlinear relationships among technological variables. In particular, IoT sensor density exhibited diminishing marginal gains beyond approximately 75 sensors per hectare.

Significant interaction effects among technologies were also observed. For instance, combining high GIS field coverage with a medium AI adoption score (50–70 points) reduced operational downtime by an additional 23.6% compared with implementing either technology independently.

Performance Metrics

Aggregate performance across all sugar mills showed:

- Productivity Index: 85.82 ± 12.37
- Cost Reduction: $14.12\% \pm 6.45\%$
- Downtime Reduction: $31.92\% \pm 14.28\%$
- Sugar Recovery Improvement: $4.86\% \pm 2.17\%$

Stratified analyses based on technology adoption quartiles demonstrated progressive improvements in productivity outcomes.

High-adoption mills (top quartile) achieved a Productivity Index of 94.71, significantly outperforming low-adoption mills (bottom quartile), which achieved a score of 76.33. The performance gap was particularly evident for:

- Downtime Reduction: 42.15% versus 19.87%
- Cost Reduction: 18.92% versus 9.45%

Mid-adoption mills demonstrated intermediate productivity levels, with Productivity Index values ranging from 82.47 to 89.16. Notably, sugar mills combining medium AI adoption with high IoT integration achieved sugar recovery improvements of 5.92%, exceeding the overall average by 21.8%.

These findings provide empirical support for the proposed technological integration framework and indicate that AI-driven approaches offer particularly substantial productivity gains.

DISCUSSION

The findings of this study contribute to both theoretical development and practical implementation in the sugar industry.

From a theoretical perspective, the results support the concept of synergistic technology implementation as stated in Yusianto et al. (2022), whereby the combined adoption of AI, GIS, and IoT technologies produces benefits greater than the sum of their individual effects. This observation is consistent with studies on digital ecosystems and industrial digital transformation by Subramaniam et al. (2019), which emphasize the value generated through interconnected technological systems.

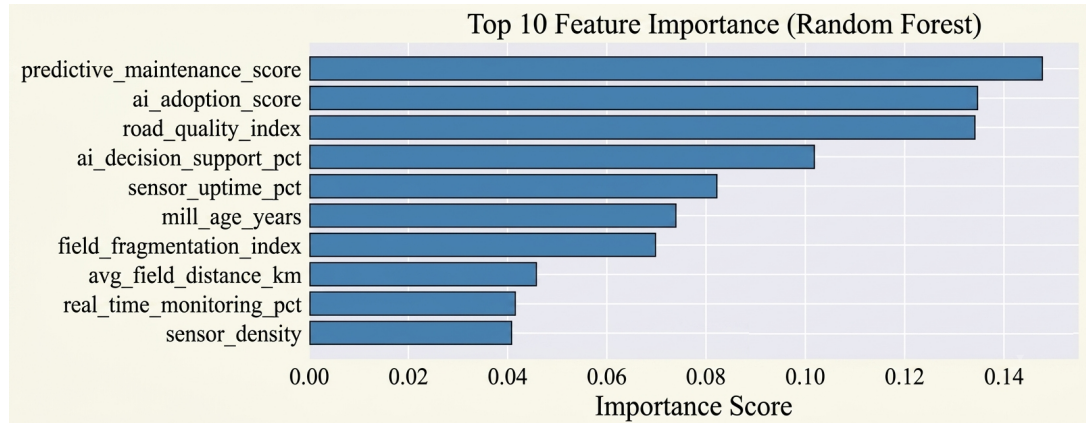
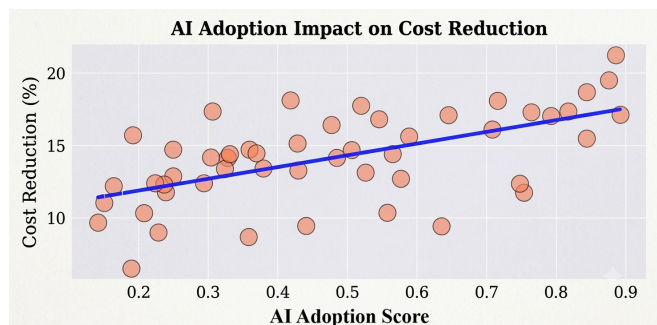
Furthermore, the diminishing returns observed for IoT sensor density suggest the existence of an optimal level of technology implementation, challenging the common assumption that increased technological investment necessarily produces proportional productivity improvements (Costa et al., 2022).

From a practical standpoint, the results provide several actionable recommendations for sugar mill operators and policymakers. The substantial contribution of Predictive Maintenance Score (14.8% feature importance) indicates that AI-enabled predictive maintenance should be prioritized for organizations seeking immediate operational improvements. Transitioning from reactive maintenance strategies to proactive, data-driven maintenance

Table 1. Correlation coefficients between technology adoption and productivity metrics.

Technology Metric	Productivity Index	Cost Reduction	Downtime Reduction	Sugar Recovery Improvement
AI Adoption Score (AAS)	0.264	0.198*	0.231	0.187*
IoT Sensor Density (ISD)	0.119*	0.154*	0.203	0.112
GIS Monitoring (GMC)	0.042 [†]	0.087	0.095	0.063

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

**Figure 1.** Top ten feature importance scores obtained from the Random Forest model.**Figure 2.** Impact of AI adoption on productivity outcomes.

practices can substantially reduce downtime and improve production efficiency (Gandhare et al., 2018).

Similarly, the observed interaction between GIS field coverage and AI adoption suggests that investments in GIS infrastructure can significantly enhance the effectiveness of existing AI systems. Policymakers and regional development agencies may therefore consider designing subsidy and incentive programs that encourage integrated technology adoption rather than isolated technological investments.

Future studies may extend the present framework in several ways:

- Conduct longitudinal analyses across multiple harvest cycles to assess the sustainability of productivity gains.
- Perform comparative studies across different countries and industrial contexts to evaluate the generalizability of the proposed framework.
- Investigate the socioeconomic implications of technology adoption, including employment patterns,

workforce skill requirements, and training needs.

- Develop hybrid modelling approaches that integrate system dynamics with machine learning techniques to better capture complex technological, environmental, and human interactions.
- Explore regional disparities in technological effectiveness to develop location-specific implementation strategies.

A deeper understanding of regional and contextual factors influencing technology adoption will facilitate the development of more effective and targeted digital transformation strategies for the sugar industry.

CONCLUSIONS

This study has revealed that the integration of AI, GIS and IoT in sugar industry has a considerable productivity gain with AI having the highest positive correlation. The results demonstrate that predictive maintenance, AI-based decision assistance and geographical observation lead to verifiable improvements to production levels and cost savings while enhancing the quantity of sugar produced. The findings validate the hypothesised benefits of the joint use of synergised technologies but also highlight the need for careful technological integration strategies especially with IoT to avoid diminishing returns. Further research needs to focus on longitudinal study of adoption of technology in real-world practices to demonstrate sustainability of these gains in productivity. Socio-economic impact on the agricultural-industrial system has yet to be deeply explored, including adaptation of workforce and differences in regions. This framework is proposed as basis for similar study as well as to offer practical im-

plications to practitioners in the industry for business optimization through appropriate use of technology.

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